

APP : Exceptions handling

Objectives :

- Define what an exception is and why it occurs.
- Explain the benefits of using Java exception handling.
- Distinguish between checked and unchecked exceptions.
- Write a simple exception handling routine to handle an exception using try-catch and try-final blocks.
- Write methods that use throws declarations for exception handling.

Activity 1 :

- 1) Test these different cases and read the messages that will be displayed on your IDE console:

1st case :

```
public class SuperHotel {  
    static int[] tab;  
  
    public void reserve() { tab[0] = 1; }  
  
    public static void main(String[] args) {  
        SuperHotel sp = new SuperHotel();  
        sp.reserve();  
    }  
}
```

2nd case :

```
import java.util.Scanner;

public class DivisionException {
    static int x = 20;
    static int y;

    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        System.out.println(" Enter a number");
        y = scanner.nextInt();
        System.out.println(x / y);
    }
}
```

Test the case if the user enters 0.

3rd case :

```
public class MethodCalling {

    public void method1() { this.method2(); }

    private void method2() { this.method1(); }

    public static void main(String[] args) {
        MethodCalling mc = new MethodCalling();
        mc.method1();
    }
}
```

4th case :

```
public class AddTable {  
  
    public static void main(String[] args) {  
        int[] array = new int[3];  
        for (int i=0;i< 4;i++){  
            array[i]= i;  
        }  
        System.out.println(array);  
    }  
}
```

- 1) Give a solution to these problems (see course)

Activity 2: The bloc *finally*

Class TestFinally.java

- 1) Compile and run the TestFinally.java file.
- 2) Modify the TestFinally.java file to handle the exception using a try-catch block in the main() method, following the instructions given in the comments.